Euro Inflation Forecasting with Bayesian VAR

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Contents

1 Introduction **2**

2 Data and Preliminary Analysis **3**
2.1 Sources and Variable Definition 3
2.2 Transformations and Data Preprocessing 4
2.3 Exploratory Analysis 4
2.3.1 Time Plots 4
2.3.2 Rolling Statistics 4
2.3.3 Time dependence: ACF and PACF 6
2.3.4 Histograms, QQ-plot and Correlation Matrix 8

3 Forecast Evaluation Design **10**

4 Forecasting Models **11**
4.1 Persistence Benchmark 11
4.2 Autoregressive Model (AR) 11
4.3 Vector Autoregression (VAR) 12
4.4 Bayesian Vector Autoregression (BVAR) 12
4.5 Prediction Intervals for the Classical Models 13
4.6 BVAR Prediction Intervals 14

5 Empirical Results **15**
5.1 Different Horizons Forecast 17

6 Conclusion **21**

1 Introduction

Inflation forecasting is a central problem in macro-economic analysis: knowing the future behavior of prices and the relative uncertainty in our beliefs is important to drive monetary policy decisions of public institutions. A low positive value for inflation is usually desirable, so facing the problem of predicting the future path of prices has a direct impact on the economy, and modifying interest rates in addition to other instruments that *ECB* dispose will directly affect inflation, and for this reason is important to develop and study models that permit to study this problem.

The objective of this work is verify if information contained in certain macroeconomic variables helps to improve the forecast accuracy obtained utilising just inflation past values and if a bayesian inference approach brings an additional benefit. Annual inflation is given by percentage variation among twelve months of *HICP*: armonized consumers price index. We will forecast monthly logarithmic inflation, and then from this reconstruct the annual inflation.

Alongside total inflation, we will consider three more variables:

- **Energy inflaction component of HICP**: This component directly contribute to HICP total index, and it may give informations about cost pressures carried to other goods and services.
- **Monthly log growth in industrial production**: This component describes a part of real economic activity and it may grasp conditions of supply and demand associated to price dynamic.
- **Euribor rate three months**: This is an indicator of monetary and financial conditions in the short term

Their contributions by the way has to be empirically evaluated and their addition does not guarantee an improvement of the models and the forecasting accuracy.

We will compare four models of increasing complexity:

- **Persistence Model**: This model assumes the future value of inflation will be the same of last observed value, and we will use it as elementary benchmark.
- **Autoregressive Model**: Now we include a linear combination of past values of inflation to model its dynamic.
- **Vector Autoregressive Model**: This model extends the AR model including also the other time-series in a vectorial framework, allowing us to study if those contains any predictive power.

- **Bayesian VAR**: Including also the other time-series increases significantly the number of parameters we have to estimate. We face this problem introducing a bayesian VAR and assigning to parameters and covariance matrix a prior distribution updated through data likelihood to give a posterior distribution.

The evaluation of the models will be done through "out-of-samples" values. An initial part of the dataset will be used to train the models: estimate parameters or evaluate distributions. We will evaluate both the accuracy of punctual estimates and reliability of forecast intervals, through error measures like MAE, RMSE, interval score and coverage. The aim is to determine whether, at what horizons and with what limits multi-variate analysis and bayesian regularization improve inflation prediction.

2 Data and Preliminary Analysis

Now we will briefly describe data and preliminary operations we do before giving them to models. The dataset is acquired through *ECB* data portal: total *HICP*, energy component of *HICP*, industrial production and three months *Euribor* rate. We will transform price and production index in order to get their logarithmic monthly inflation (and logarithmic industrial growth), according to:

$$\pi_t = \ln P_t - \ln P_{t-1} \tag{1}$$

2.1 Sources and Variable Definition

Feature	**Source**	**Description**
HICP Total	Eurostat: ECB Data Portal	Harmonized Index of Consumer Prices, not seasonally adjusted, for the euro area with a variable composition.
HICP Energy	Eurostat: ECB Data Portal	Energy component of the HICP, not seasonally adjusted, for the euro area with variable composition.
Industrial Production	Eurostat: ECB Data Portal	Industrial Production Index, calendar and seasonally adjusted, for the Eurozone EA20 fixed composition.
Euribor 3-months:	ECB Data Portal, dataset Financial Market	Monthly average of the three-month nominal Euribor rate, expressed as an annual percentage.

Table 1: Original monthly series used in the analysis and related sources.

2.2 Transformations and Data Preprocessing

We acquired monthly data for our four time-series in a period that goes from January 2000 to December 2025.

Each series contains 312 observations and there are no missing values, so we did not need to imputate any observation. After a preliminary data exploration, there is no need to eliminate any outlier too.

We transformed the total HICP index, Energy HICP and industrial production according to the following relation: let P_t be the value of an index at month t, then we consider the following transformation:

$$\pi_t = \ln P_t - \ln P_{t-1} \tag{2}$$

So now the time series became the logarithmic variation with respect to the previous month: for both HICP indexes we get respectively monthly logarithmic inflation and monthly energy inflation, and about total industrial production now we get logarithmic industrial monthly growth rate. We did not transform the Euribor time series. The variables are aligned by month and arranged in the following order:

$$\mathbf{z}_t = \begin{pmatrix} \pi_t \\ \pi_t^E \\ g_t^{IP} \\ i_t^{3M} \end{pmatrix} \tag{3}$$

where π_t is total inflation, π_t^E energy inflation, g_t^{IP} industrial growth and i_t^{3M} three months Euribor.

2.3 Exploratory Analysis

This preliminary exploratory data analysis is made on the same time interval of train set: here we want to describe the dynamic of our four features and identify important characteristics of the time series.

2.3.1 Time Plots

The time plots, Figure 1, reveal different dynamics across the four variables in the initial sample, from February 2000 to July 2015. Monthly total inflation exhibits recurring fluctuations whose amplitude increases towards the end of the sample. Energy inflation displays larger movements, including pronounced negative observations in 2008–2009. Industrial production growth also shows a substantial contraction around this period, followed by a recovery. Euribor evolves more smoothly, with persistent movements between high- and low-rate periods and a marked decline after 2008.

2.3.2 Rolling Statistics

The trailing twelve-month statistics showed in Figure 2 reveal changes in both location and dispersion. Total inflation's rolling mean peaks around 2008 and



Figure 1: Monthly trends in total inflation, energy inflation, industrial growth, and the three-month Euribor over the period considered in train set. The panels use different vertical scales.

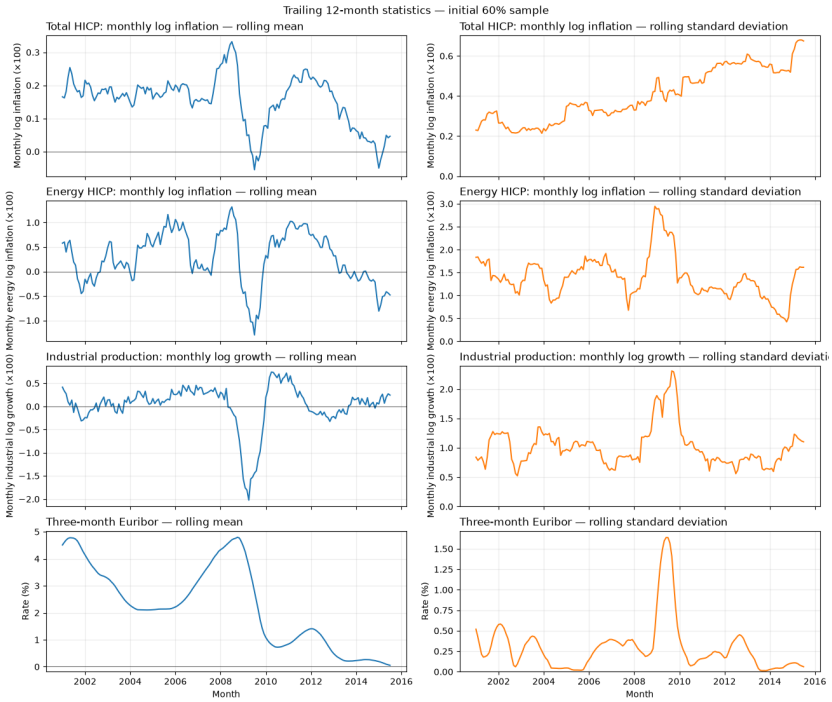


Figure 2: Mean and standard deviation of the four variables over 12-month time windows. Each point refers to the final month of the window.

subsequently declines, while its rolling standard deviation generally increases across the sample. Energy inflation and industrial production growth display pronounced declines in their rolling means and peaks in dispersion around 2008–2009. For Euribor, elevated rolling dispersion accompanies the rapid transition to lower rates.

2.3.3 Time dependence: ACF and PACF

Total inflation displays a pronounced seasonal pattern as we can see in Figure 3: the ACF has positive peaks at multiples of six months, particularly at lags 12, 24 and 36. The PACF also shows substantial coefficients at lags six and twelve. Energy inflation exhibits positive first-order dependence, with weaker correlations at most longer lags. Industrial production growth shows positive correlations at lags two and three. Euribor has a slowly declining ACF and a dominant first-lag PACF, indicating strong persistence in levels. These patterns help characterise temporal dependence, but do not by themselves identify the optimal forecasting specification.

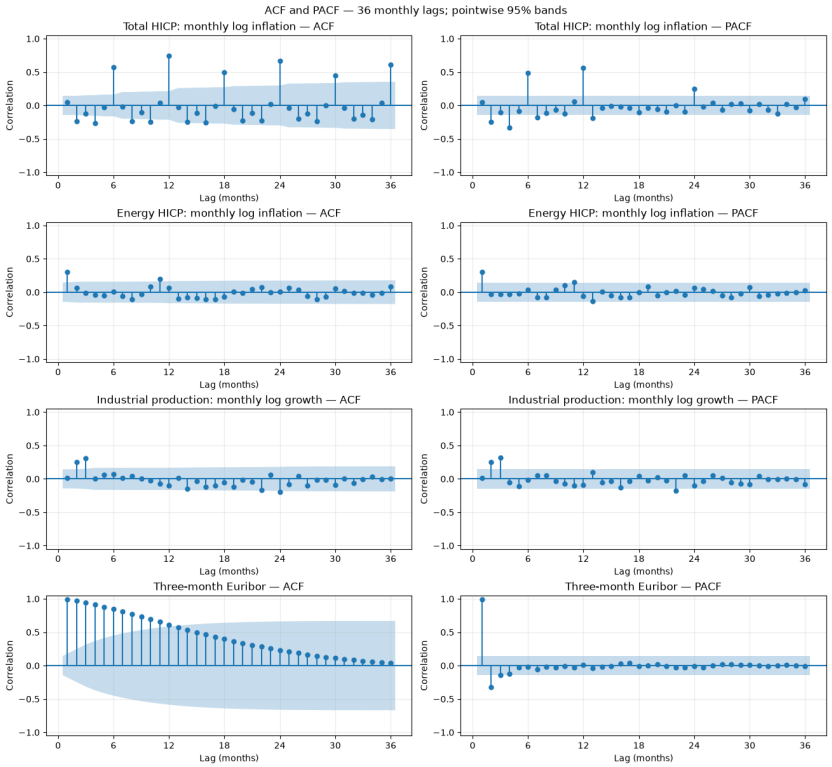


Figure 3: Autocorrelation functions (ACF) and partial autocorrelation functions (PACF) of the four variables up to 36 months. The shaded areas represent pointwise 95% confidence bands.

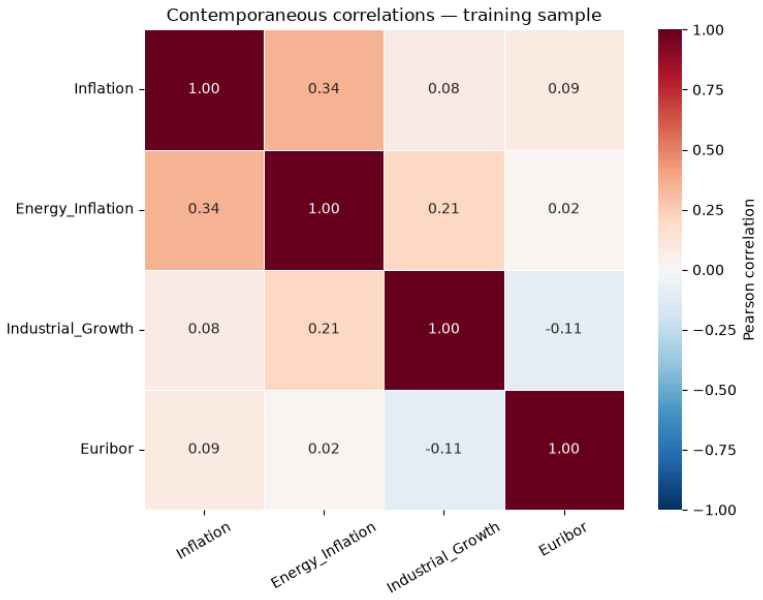


Figure 4: Contemporaneous Pearson correlations between the four model variables in the initial training sample.

2.3.4 Histograms, QQ-plot and Correlation Matrix

The strongest contemporaneous association is between total and energy inflation, with a Pearson correlation of approximately 0.34. Total inflation has much weaker associations with industrial production growth and Euribor, at 0.08 and 0.09, respectively. Energy inflation and industrial production growth have a correlation of 0.21, while industrial growth and Euribor exhibit a weak negative correlation of -0.11. These coefficients describe same-month linear associations, and are showed in Figure 4.

The distributions of the three monthly log differences are broadly concentrated around their centres, Figure 5, but their tails depart from the normal reference. Total inflation shows deviations in both tails, whereas energy inflation and industrial production growth display particularly pronounced negative extremes. Euribor has an irregular distribution with concentrations at different rate levels and substantial departures from the normal Q–Q line.

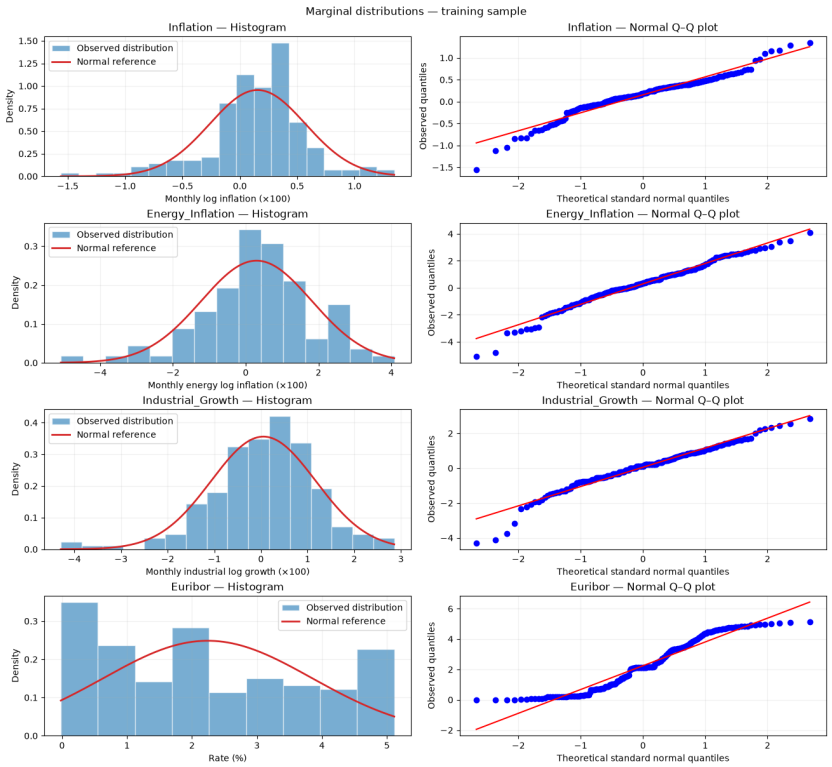


Figure 5: Histograms with normal reference densities and normal Q–Q plots of the four model variables in the initial training sample.

3 Forecast Evaluation Design

The forecasting procedure is made using an expanding window technique and a fixed horizon of 1,3,6,12 months. Our dataset contains monthly observation going from February 2000 to December 2025, for a total number of 311 values. We begin splitting the full dataset in two parts: first 60% of it, will constitute the beginning train dataset, then the remaining 40% is the evaluation period. Then for each target month T, the forecast origin is $t = T - HORIZON$, and each model is trained every step with only the information available until t and it generates forecasts of the following $HORIZON$ months.

Let $\{\boldsymbol{X}_t\}_{t=0}^{t=t_{fin}}$ be our full dataset, and $t_{split} = \lfloor 0.6 \cdot t_{fin} \rfloor$ the month where we split the dataset, the forecasting target will be $T = t_{split}, t_{split} + 1, \ldots, t_{fin}$. For each step, we set $t = T - HORIZON$, we train again our models and we let them forecast the next $HORIZON$ months.

At each origin, the models forecast monthly log inflation. Annual log inflation is then reconstructed using the identity

$$a_T = \log P_T - \log P_{T-12} = \sum_{j=0}^{11} \pi_{T-j}, \qquad \pi_t = \log P_t - \log P_{t-1} \tag{4}$$

where P_t denotes the total HICP. At a six-month horizon, this sum contains six monthly observations already known at the forecast origin and six future monthly values. The annual forecast is therefore

$$\widehat{a}_{t+6|t} = \underbrace{\sum_{j=0}^{5} \pi_{t-j}}_{\text{observed component}} + \underbrace{\sum_{h=1}^{6} \widehat{\pi}_{t+h|t}}_{\text{forecast component}} \tag{5}$$

All six monthly forecasts in the second term are generated from the same origin, without incorporating observations after t. For the BVAR, the same reconstruction is applied to each simulated predictive trajectory, and the annual point forecast is obtained by averaging the reconstructed values. Results are reported as $100a_T$.

For example, at an origin of January 2020, the target month is July 2020. Annual inflation at that target measures the log change in the price index between July 2019 and July 2020. Its reconstruction combines the observed monthly changes from August 2019 to January 2020 with forecasts for February to July 2020.

The origin then advances by one month. The newly observed data are added to the estimation sample, the models are re-estimated, and a new six-month forecast path is generated.

This design represents a sequence of forecasting decisions in which the information set is updated over time.

4 Forecasting Models

All four models follow the expanding-window forecasting design described above. At each origin t, they generate a path of monthly inflation forecasts $\{\widehat{\pi}_{t+h|t}\}_{h=1}^{HORIZON}$, using only observations up to that origin. The six-month-ahead monthly forecast is the final value of this path, whereas the annual forecast combines all six predicted monthly values with the six observed monthly changes from $t-5$ to t.

4.1 Persistence Benchmark

The persistence model provides a simple benchmark by assuming that monthly inflation remains equal to its most recently observed value throughout the forecast horizon:

$$\widehat{\pi}_{t+h|t} = \pi_t, \qquad h = 1, \ldots, 6 \tag{6}$$

The corresponding annual forecast is

$$\widehat{a}_{t+6|t} = \sum_{j=0}^{5} \pi_{t-j} + 6\pi_t \tag{7}$$

Thus, persistence is imposed on monthly inflation, rather than directly on annual inflation. The annual forecast can differ from the latest observed annual value because the composition of its twelve-month window changes.

This benchmark does not require estimation of coefficients for its point forecasts. At each new origin, the constant forecast path is updated using the latest monthly observation.

4.2 Autoregressive Model (AR)

The autoregressive model uses the previous six monthly inflation observations to describe current inflation:

$$\pi_s = c + \sum_{j=1}^{6} \phi_j \pi_{s-j} + \varepsilon_s \tag{8}$$

where c is an intercept and ε_s is a zero-mean innovation. The lag order is fixed at six, and the coefficients are estimated by ordinary least squares at each forecast origin using the expanding estimation sample.

Multi-step forecasts are generated recursively:

$$\widehat{\pi}_{t+h|t} = \widehat{c}_t + \sum_{j=1}^{6} \widehat{\phi}_{j,t} \begin{cases} \pi_{t+h-j}, & h-j \leq 0 \\ \widehat{\pi}_{t+h-j|t}, & h-j > 0 \end{cases} \tag{9}$$

Observed values are used whenever the required lag falls at or before the origin; otherwise, the model uses its own previously generated forecasts. Future

innovations are set to their assumed conditional mean of zero. The estimated coefficients remain fixed throughout each six-month forecast path and are re-estimated when the origin advances.

The annual forecast is obtained by summing the six monthly forecasts and adding the observed component, as defined in the forecasting design.

4.3 Vector Autoregression (VAR)

The vector autoregressive model extends this framework by jointly modelling total inflation, energy inflation, industrial production growth, and the three-month Euribor:

$$\mathbf{z}_s = \begin{pmatrix} \pi_s \\ \pi_s^E \\ g_s^{IP} \\ i_s^{3M} \end{pmatrix} \tag{10}$$

Its specification is

$$\mathbf{z}_s = \mathbf{c} + \sum_{j=1}^{p} A_j \mathbf{z}_{s-j} + \boldsymbol{\varepsilon}_s \tag{11}$$

where $\mathbf{c}$ is an intercept vector, A_j are coefficient matrices, and $\boldsymbol{\varepsilon}_s$ is a zero-mean innovation vector with covariance matrix Σ. Now each variable will therefore depend on its own lags and on the lags of the other variables.

At each origin, the lag order is selected by minimising the Akaike information criterion over $p \in \{0, \ldots, 8\}$. The coefficients are estimated by ordinary least squares.

Forecasts are generated recursively for all four variables. Beyond the origin, predicted values of the explanatory variables are used in place of their future observations. Both the selected order and the estimated coefficients remain fixed within each forecast path and are updated at the next origin.

The first component of each predicted vector provides the monthly total-inflation forecast:

$$\widehat{\pi}_{t+h|t} = \mathbf{e}_1^\top \widehat{\mathbf{z}}_{t+h|t}, \qquad \mathbf{e}_1 = (1, 0, 0, 0)^\top \tag{12}$$

These six monthly forecasts are then combined with the six known monthly changes to reconstruct annual inflation. The comparison with the univariate models assesses whether including the additional macroeconomic variables improves forecasting performance under the chosen specifications.

4.4 Bayesian Vector Autoregression (BVAR)

The Bayesian VAR jointly models the four variables using twelve lags and an intercept, without a deterministic trend. Each variable is standardised using its mean and sample standard deviation computed from observations available at

the first forecast origin. The model is

$$\widetilde{\mathbf{z}}_s = \mathbf{c} + \sum_{\ell=1}^{p=12} A_\ell \widetilde{\mathbf{z}}_{s-\ell} + \boldsymbol{\varepsilon}_s, \qquad \boldsymbol{\varepsilon}_s \mid \Sigma \overset{\text{iid}}{\sim} \mathcal{N}_4(\mathbf{0}, \Sigma) \tag{13}$$

A matrix-normal–inverse-Wishart conjugate prior is assigned to the coefficient matrix B and innovation covariance Σ:

$$\Sigma \sim IW(I_4, 6), \qquad B \mid \Sigma \sim MN(M, V, \Sigma) \tag{14}$$

The prior coefficient means are zero. Minnesota-style regularisation is specified through MATLAB's `SelfLag = 0.20` and `Decay = 2` options, encouraging small autoregressive coefficients and stronger shrinkage at longer lags. In the conjugate specification, the same tightness setting applies to own-variable and cross-variable lags; the intercepts receive a weakly informative prior. At each monthly origin, the joint posterior distribution is updated using the expanding estimation sample and the same fixed prior. The lag order and prior hyperparameters are not re-selected during evaluation.

4.5 Prediction Intervals for the Classical Models

Prediction intervals are constructed at each forecast origin, treating the estimated parameters as fixed. Since the observed component of annual inflation is already known, uncertainty arises exclusively from the sum of future monthly forecasted inflation values. Denoting the monthly forecast error by $e_{t,h} = \pi_{t+h} - \widehat{\pi}_{t+h|t}$, we consider the variance

$$V_{t,H} = \operatorname{Var}_t \left(\sum_{h=1}^{H} e_{t,h} \right) \tag{15}$$

This calculation accounts for dependence between errors at different horizons, which share some of the same future innovations.

For the AR and VAR models, future innovations are propagated through the estimated autoregressive dynamics. In the VAR, the propagation matrices satisfy

$$\Psi_0 = I, \qquad \Psi_j = \sum_{\ell=1}^{\min(p,j)} \widehat{A}_\ell \Psi_{j-\ell} \tag{16}$$

Defining $C_j = \sum_{k=0}^{j} \Psi_k$, the variance of the annual forecast error is

$$V_{t,H} = \sum_{j=0}^{H-1} \mathbf{e}_1^\top C_j \widehat{\Sigma} C_j^\top \mathbf{e}_1 \tag{17}$$

where $\mathbf{e}_1$ selects total inflation and $\widehat{\Sigma}$ is the estimated innovation covariance matrix. The cumulative matrices capture the contribution of each innovation to

several monthly terms in the annual sum. The AR calculation is the univariate version of the same expression.

For persistence, uncertainty is specified through a random walk without drift for monthly inflation: $\pi_s = \pi_{s-1} + u_s$. Its innovation variance is estimated as the mean squared first difference of monthly inflation within the current estimation window. Since each innovation affects all subsequent monthly values, the annual forecast-error variance becomes

$$V_{t,H}^{P} = \widehat{\sigma}_u^2 \sum_{k=1}^{H} k^2 \tag{18}$$

Assuming Gaussian innovations, the annual forecast error is Gaussian conditional on fixed parameters. Using the estimated variance, the central prediction interval with nominal coverage $1 - \alpha$ is therefore

$$\left[\widehat{a}_{t+H|t} - z_{1-\alpha/2}\sqrt{V_{t,H}}, \quad \widehat{a}_{t+H|t} + z_{1-\alpha/2}\sqrt{V_{t,H}}\right] \tag{19}$$

where z_q is the standard quantile of normal distribution. We report intervals at the 80% and 95% levels.

4.6 BVAR Prediction Intervals

At each forecast origin, we generate 20,000 trajectories from the posterior predictive distribution. For each trajectory, a coefficient matrix and innovation covariance are jointly drawn from the posterior and held fixed throughout the six-month forecast horizon. New Gaussian innovations are generated at each step, and all four variables are simulated recursively. The simulated total-inflation values are then transformed back to their original monthly log units.

For trajectory s, annual inflation is reconstructed as

$$a_{t+6}^{(s)} = \sum_{j=0}^{5} \pi_{t-j} + \sum_{h=1}^{6} \pi_{t+h}^{(s)} \tag{20}$$

The annual point forecast is the mean of these reconstructed values. Central prediction intervals are obtained directly from their empirical quantiles:

$$PI_{80\%} = [q_{0.10}, q_{0.90}], \qquad PI_{95\%} = [q_{0.025}, q_{0.975}] \tag{21}$$

This procedure incorporates uncertainty in the coefficients, innovation covariance and future innovations, while preserving dependence between monthly values within each trajectory. Although innovations are conditionally Gaussian, no Gaussian approximation is imposed on the final predictive distribution. The intervals remain conditional on the chosen model specification and fixed hyperparameters.

5 Empirical Results

Forecasting performance is evaluated using three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Interval Score (IS). All metrics refer to six-month-ahead forecasts of annual log inflation, expressed on the scale $100a_T$.

Let N denote the number of evaluation targets, y_n the observed value, and $\hat{y}_n$ its point forecast. The MAE is

$$\text{MAE} = \frac{1}{N}\sum_{n=1}^{N}|y_n - \hat{y}_n|,$$

while the RMSE is

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}(y_n - \hat{y}_n)^2}.$$

Both measure point forecast accuracy, with the RMSE placing greater weight on large errors.

For a prediction interval $[L_n, U_n]$ with nominal coverage $1-\alpha$, the interval score is

$$\text{IS}_{\alpha,n} = (U_n - L_n) + \frac{2}{\alpha}\max(L_n - y_n, 0) + \frac{2}{\alpha}\max(y_n - U_n, 0).$$

The reported score is its average across evaluation targets:

$$\overline{\text{IS}}_\alpha = \frac{1}{N}\sum_{n=1}^{N}\text{IS}_{\alpha,n}.$$

This metric combines interval width with penalties for observations falling outside the interval. We calculate it separately for $\alpha = 0.20$ and $\alpha = 0.05$, corresponding to the 80% and 95% prediction intervals. Lower values indicate better performance for all three metrics; interval scores are compared across models at the same nominal coverage level.

Model	Horizon [months]	Level (%)	MAE	RMSE	Coverage	Interval score
Persistence	6	80	2.3380	3.1012	0.968	15.3306
Persistence	6	95	2.3380	3.1012	1.000	22.5261
AR	6	80	1.0755	**1.4067**	0.592	5.4387
AR	6	**95**	1.0755	**1.4067**	0.808	**10.8490**
VAR	6	80	1.1319	1.6051	0.552	6.3172
VAR	6	95	1.1319	1.6051	0.776	13.4625
BVAR	6	**80**	**0.9707**	1.4641	0.736	**5.4113**
BVAR	6	95	**0.9707**	1.4641	0.872	11.2802

Table 2: Forecast accuracy and prediction interval performance for annual log inflation at a six-month horizon. MAE, RMSE, and interval score are expressed in percentage points of annual log inflation. Coverage is reported as a proportion.

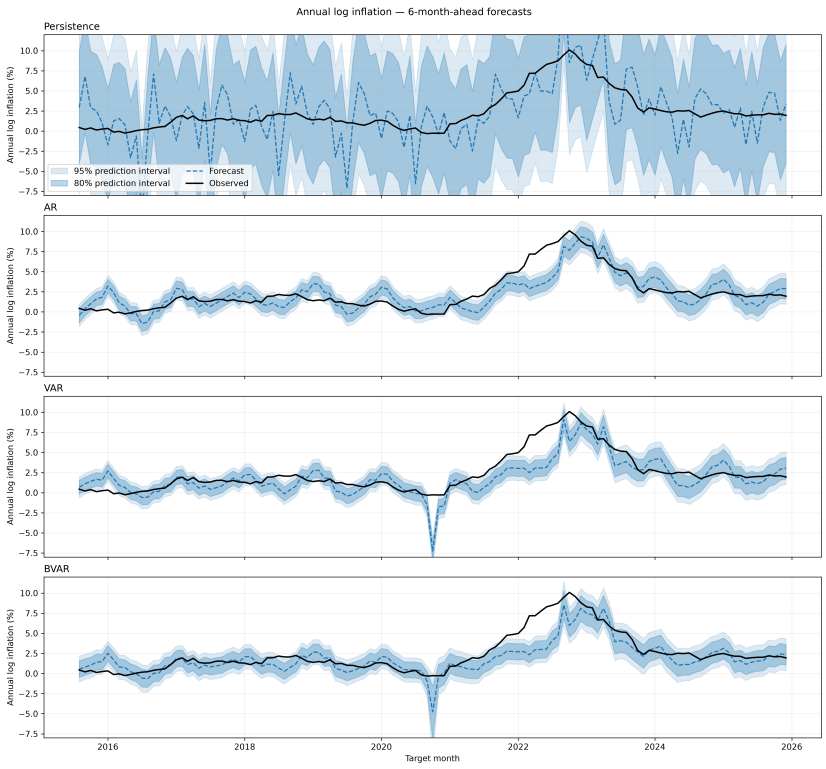


Figure 6: Annual inflation forecasting with 6 months horizon comparison among our different statistical models. The order, from top to bottom is the following: Persistence, AR, VAR, BVAR

5.1 Different Horizons Forecast

Model	Horizon	Level	MAE	RMSE	Coverage	Interval Score
Persistence	1	80	0.5035	0.6358	0.8400	2.2858
Persistence	1	95	0.5035	0.6358	0.9200	3.2899
AR	1	80	0.3408	0.4613	0.7440	1.6600
AR	1	95	0.3408	0.4613	0.8880	2.7097
VAR	1	80	0.4093	0.5141	0.6480	1.9370
VAR	1	95	0.4093	0.5141	0.8160	3.2367
BVAR	1	80	**0.3325**	**0.4485**	0.7760	**1.6170**
BVAR	1	95	**0.3325**	**0.4485**	0.8960	**2.5511**

Model	Horizon	Level	MAE	RMSE	Coverage	Interval Score
Persistence	3	80	1.2897	1.7453	0.9040	6.9311
Persistence	3	95	1.2897	1.7453	0.9520	10.0395
AR	3	80	0.7547	0.9676	0.6080	3.6408
AR	3	95	0.7547	0.9676	0.8480	**6.0552**
VAR	3	80	0.8936	1.1087	0.4960	4.4613
VAR	3	95	0.8936	1.1087	0.7600	8.2354
BVAR	3	80	**0.7095**	**0.9566**	0.7200	**3.4709**
BVAR	3	95	**0.7095**	**0.9566**	0.8720	6.0675

Model	Horizon	Level	MAE	RMSE	Coverage	Interval Score
Persistence	12	80	4.6247	6.2427	0.9760	39.5046
Persistence	12	95	4.6247	6.2427	1.0000	59.9257
AR	12	80	**1.4587**	**2.3428**	0.7200	**9.4186**
AR	12	95	**1.4587**	**2.3428**	0.8480	**22.7789**
VAR	12	80	1.8092	3.2464	0.6880	12.2228
VAR	12	95	1.8092	3.2464	0.8080	33.3484
BVAR	12	80	1.6228	2.7111	0.7440	10.2718
BVAR	12	95	1.6228	2.7111	0.8400	25.2286

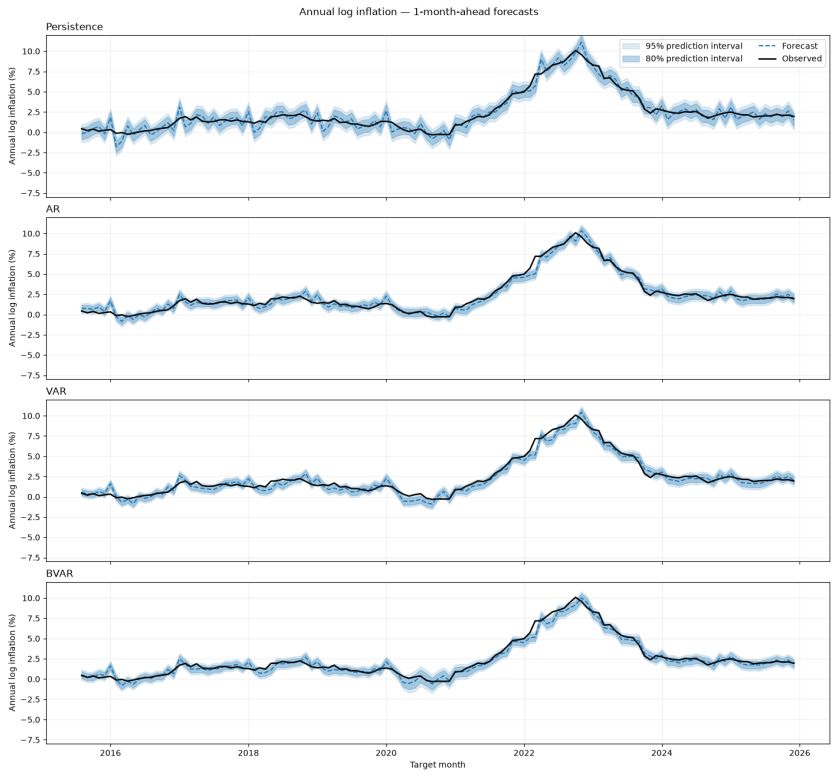


Figure 7: Annual inflation forecasting with 1 month horizon comparison among our different statistical models. The order, from top to bottom is the following: Persistence, AR, VAR, BVAR

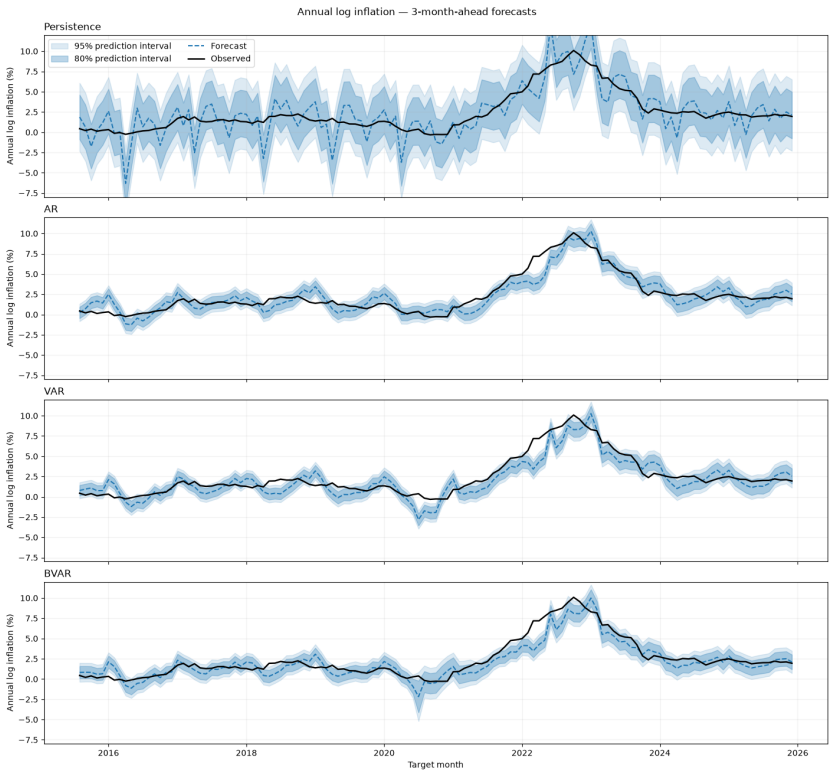


Figure 8: Annual inflation forecasting with 3 months horizon comparison among our different statistical models. The order, from top to bottom is the following: Persistence, AR, VAR, BVAR

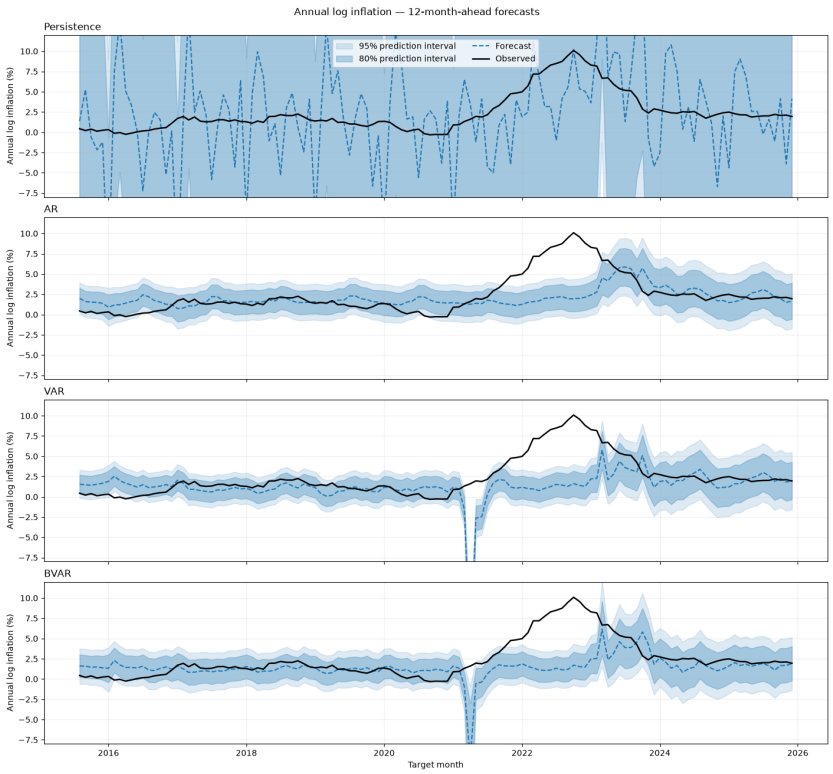


Figure 9: Annual inflation forecasting with 12 months horizon comparison among our different statistical models. The order, from top to bottom is the following: Persistence, AR, VAR, BVAR

6 Conclusion

This study examined whether additional macroeconomic variables and Bayesian estimation improve forecasts of annual log inflation in the euro area at horizons of one, three, six and twelve months. The results show that relative performance depends on both the forecast horizon and the evaluation criterion. The BVAR achieves the lowest MAE and RMSE at one and three months. At six months, it retains the lowest MAE, while the univariate AR achieves the lowest RMSE. At twelve months, the AR performs best on both point forecast metrics. The classical VAR performs worse than the AR and BVAR on both metrics at every horizon considered.

Prediction interval performance also varies with the horizon. The BVAR achieves the lowest interval score at both coverage levels at one month and at the 80% level at three and six months. The AR has a marginally lower 95% interval score at three months and a clearer advantage at six months. At twelve months, the AR achieves the lowest interval score at both levels. BVAR coverage is consistently closer to nominal levels than VAR coverage and generally closer than AR coverage, although the AR has slightly better 95% coverage at twelve months. Nevertheless, all three estimated models remain under-calibrated at both levels across the horizons considered.

Persistence has the largest point forecast errors at every horizon. As the horizon increases, its intervals become very wide and generally exceed nominal coverage, producing high interval scores. Its high coverage therefore does not imply a more informative representation of predictive uncertainty. Point forecast errors generally increase with the horizon, as a larger share of the twelve-month inflation measure must be forecast rather than reconstructed from observed values.

Overall, the Bayesian procedure is particularly competitive at shorter horizons, while the simpler AR performs better at twelve months. These findings provide no uniform advantage from multivariate modelling. Since the procedures differ in lag orders and treatment of parameter uncertainty, the comparison does not isolate the effect of Bayesian shrinkage alone. Furthermore, the reported differences are descriptive and do not establish statistical superiority. Forecasting procedures should therefore be assessed jointly through point accuracy, interval performance and the horizon relevant to their intended use.